Disaggregation is happening; OEM layers are getting middleware and many more things. So if we split the monolithic architecture, systems integrators are needed to play the role, leading to a lot more challenges.

"But the driving factor in open source is the cost of the solution and the diverse supply chain. They are pretty interrelated. I use open source and it will be cheaper, because there is nobody to charge me the royalty or the fees. We are not saying that the source is going to come cheap. We are not saying by any means it is free or it is not free. Open source is not free — it is a very common term that we can clarify all the time," explains Gangey.

Open source is the method of getting in more players and fostering innovation. Once you have the vendor lock-in, whatever the requirement, you are dependent on a set of engineers who are catering to 15 different sets of customers across the world having 15 different requirements.

Gangey says, "If my solution doesn't get priority in my partner's ecosystem, I am really stuck and will not be able to service my customer. So fundamentally, when you disaggregate, you unlock and remove this lock-in, opening up for more innovation. You deploy the features faster and are able to monetise."

Cost can be brought down in one way or the other — in the form of efficiency in deployment, with optics or by way of capital expenditures. "It is not as simple as it sounds, but it is evident that whenever you open up things, it basically brings down the overall total cost of ownership (TCO)," he adds.

Gajare says, "The intelligent programmable infra and the architectural building blocks with open source, ONAP, policy, design, creation, dashboarding, external APIs, orchestration, networking, VNS, and infra monitoring – all of this is now a bundled offering."

While he points out innumerable use cases of ONAP that can design, create, orchestrate and automate everything in 5G, he also tells telco operators to be aware that code repositories are no longer locked in. It's not just about 10,000 people contributing, but about how the whole wheel aligns to the common cause of the use case.

"I think I can see that balance. A lot of vendors and telcos are working with Telstra, COX, Orange, Charter, Bell, AT&T, for example, and have evolved from lab ONAP networks to ONAP reference architectures. And that epochal shift is a very good science," adds Gajare.

Specifics on leveraging 5G from core to edge

Various workloads have a variety of use cases and there is sometimes a doubt about how end-to-end service orchestration can happen. These could be wireless networks or CNF/VNF combinations. Going into the specifics of slice orchestration, it can be the core or access. Even in the case of access, there has to be some thought on how RAN automation is going to happen and how to support the real-time or non-real time scenarios.

"When we talk about the services being offered, there are other digital players too apart from the telco specific services. So there should be an idea about how I am going to do the end-to-end telco-cloud orchestration. Bringing in orchestrations and adding edge factors into them will still need a lot of thinking on how orchestration happens from the edge layer and when moving towards the CPE side," explains Gnanapriya.

Based on the kind of requirements, different kinds of flavours can be easily addressed in an automated dynamic orchestration through these platforms. It gives enormous facilities and capabilities where these solutions can be leveraged. In the Indian context, not just the operational efficiencies, but also from a cost perspective, bringing all these concepts together plays a vital role.

Smart cities stay smart with open source

The smart city concept is gaining momentum with all products coming under a digital shadow. It involves a functional and structural improvement of existing cities by capitalising on information and communications technology to increase the city's sustainable growth, while ensuring enhanced quality of life for its citizens.

Dr Gopal points out to one of the open source platforms, India Urban Data Exchange, that's being deployed across the country for sharing data. If the solutions that smart cities are deploying are looked at, there is a layer of software — a middleware layer, which for the most part is non-differentiated. It allows you to find data and provide access to controls, he explains.

"Companies work together to create a common open source platform. This has a significant role in the smart cities space. Everyone understands, and can deploy and support, but one has to focus on creating value on top," says Dr Gopal.

Bridge gaps with an organisational construct

India has a pretty ambitious initiative of building 100 smart cities, and this project has been going on for about six years. However, it has not moved quickly and the results have not been as dramatic as expected initially.

"It's because a lot of data — records, videos, etc — are being collected in these smart cities, but they sit right at the silos or closed proprietary platforms without any well-architectured interfaces," says Dr Gopal.

It is very difficult for someone who has an innovative idea to get access to this data or even understand what data is available. Just suppose you want to build some kind of emergency response system within a city and would like to take data from the police and fire departments, as well as hospital services. Right now, that is not possible in these cities as these departments are working on completely different systems that have no correlation. Common application programming interfaces (APIs) help in bridging this gap, creating a platform that connects the systems together and brings the available data into a common format.